Entropy and the Central Limit Theorem (CLT)

Liyan Gong; Shujun Xiong

Introduction

- Consider S_n, which is the standardized sum of i.i.d. random variables with finite variance.

$$S_n = \frac{1}{n}\sum_{i=1}^{n} X_i, X_i \sim \mathcal{X}, 1 \leq i \leq n.$$

- Regular CLT shows that $\lim_{n\to\infty} S_n \to \Phi$, where $\Phi = \mathcal{N}(\mathbb{E}(\mathcal{X}), \mathbb{V}(\mathcal{X}))$ (i.e. convergence in distribution to the **associated Gaussian distribution**)
- Define relative entropy

$$D(X) := D_{KL}(X||\Phi_X), \Phi_X := \text{associated Gaussian of } X.$$

- This paper shows convergence of $D_n := D(S_n) \to 0$. In other words, there is CLT-like convergence in relative entropy, a stronger result.

Main Result

Theorem. Let S_n be the standardized sum. Then the sequence $n\,D(S_n)$ is subadditive. In particular,

$$D(S_{2n}) < D(S_n).$$

Furthermore, the relative entropy converges to zero:

$$\lim_{n\to\infty} D(S_n) = 0$$

if and only if $D(S_n)$ is finite for some n. Equivalently, the entropy $H(S_n)$ converges to the normal entropy

$$\frac{1}{2}\log(2\pi e\sigma^2),$$

provided the entropy is finite for some n.

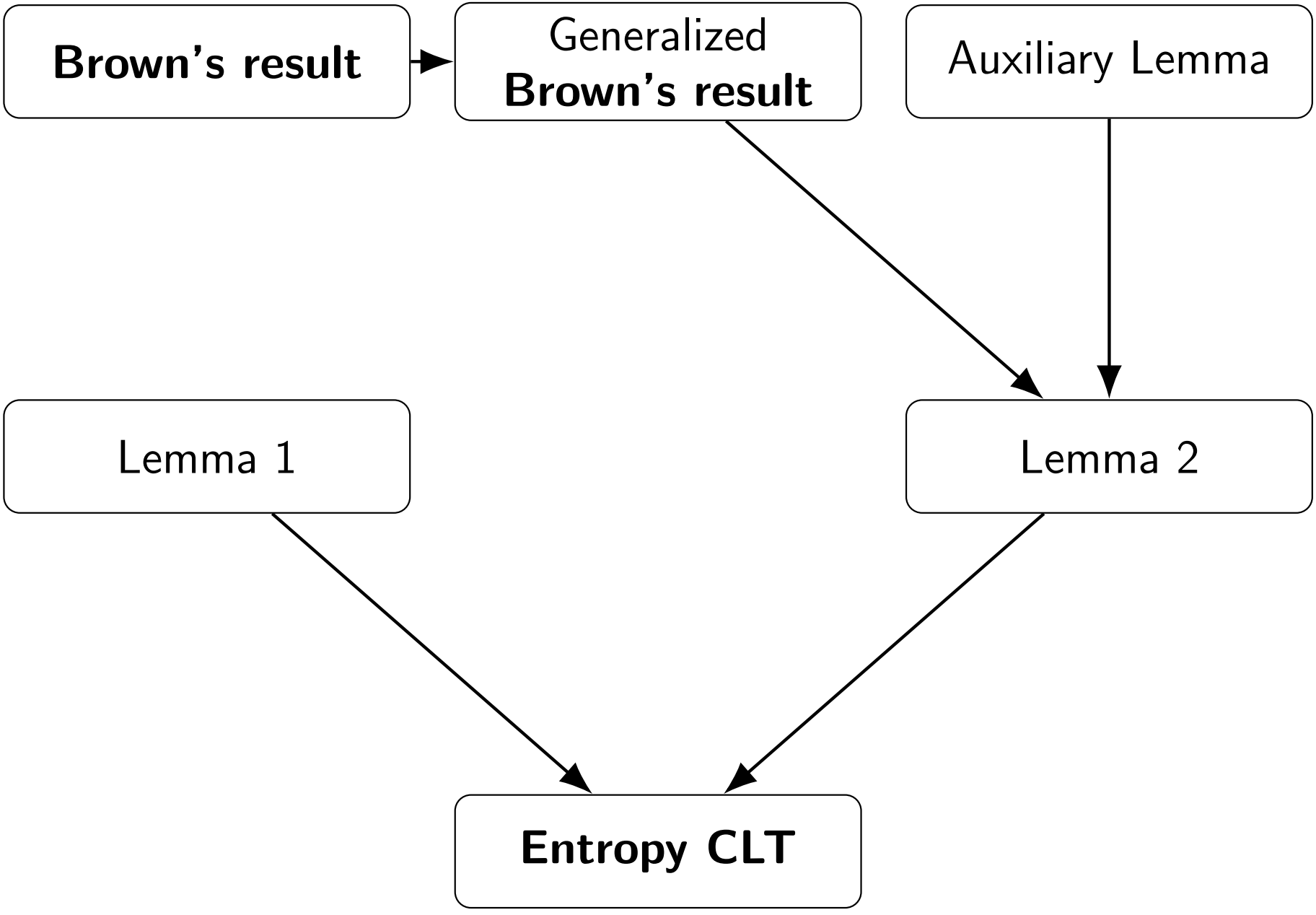
Motivations

- Previous work in Rényi (1970): showed convergence of $\int f_n \log \frac{f_n}{\phi} \to 0$ for smoothed densities. Linnik (1959) proved this only for truncated and Gaussian-smoothed densities.
- Brown (1982) derived a well-motivated conjecture that the Fisher information of the standardized sum is equal 0.
- de Bruijn's identity from class showed a relationship between Fisher information and entropy, and we hope to use this well.
- **Our result:** Proof of Brown's conjecture implies $D_n \to 0$ holds without any smoothness assumptions on f_n.(if D_n is finite at some point).

Overview of proof strategy

- We aim to prove our main result, **Entropic CLT**.
- **Entropic CLT** relies on two key intermediate steps: Lemma 1 and Lemma 2.
- To obtain Lemma 2, we first generalize **Brown's result**. Then, we use this generalized version and an auxiliary lemma to prove Lemma 2.

Proof Structure



Use Lemma 1 to obtain the relationship of $D(X)$ and $J(Y_t)$

Let X be a random variable with finite variance:
$Z \sim \mathcal{N}(\mathbb{E}(X), \mathbb{V}(X))$ and X, Z are independent. Then,

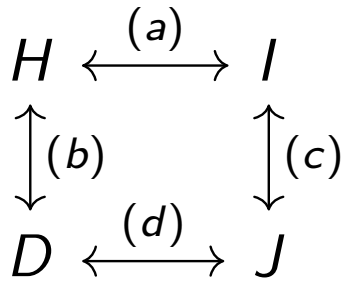
$$\boxed{D(X) = \int_0^1 J(Y_t)\frac{dt}{2t}, Y_t = \sqrt{t}X + \sqrt{1-t}Z.}$$

This result follows from a change of variables after de Bruijn's identity

$$\frac{d}{d\tau}H(X + Z_\tau) = \frac{1}{2}I(X + Z_\tau).$$

In the interest of time, we will forgo the details, but we will give the proof strategy of Lemma 1.

Proof strategy of Lemma 1



We construct **standardized Fisher Information**

$$J(X) = \sigma^2 \mathbb{E}(\rho(X) - \rho_\phi(X))^2$$

for score function of X and $\Phi(X)$, whose density functions are $\rho = \frac{f'}{f}$ and $\rho_\phi = \frac{\phi'}{\phi}$.
Note that in this talk we will be using I for Fisher information, rather than mutual information.

Proof strategy of Lemma 1

A natural set of relations emerges between entropy and Fisher information with *relative* entropy and *relative* Fisher information

$$\begin{array}{ccc} H & \overset{(a)}{\longleftrightarrow} & I \\ \updownarrow (b) & & \updownarrow (c) \\ D & \overset{(d)}{\longleftrightarrow} & J \end{array}$$

- (a) is given by de Bruijn's identity: for standard normal Z and $t > 0$, $\frac{d}{dt} H(X + \sqrt{t}Z) = \frac{1}{2} I(X + \sqrt{t}Z)$.
- (b) is given by connection: $H(X) = \frac{1}{2} \log(2\pi e \sigma^2) - D(X)$.
- (c) is given by connection: $I(X) = \frac{J+1}{\mathbb{V}(X)}$.
- (d) is given by Lemma 1 of our proof: $D(X) = \int_0^1 J(\sqrt{t}X + \sqrt{1-t}Z)\frac{dt}{2t}$, where Z is the standard normal Gaussian independent of X.

Remark

Brown (1982) demonstrated the existence of the limit

$$\lim_{n\to\infty} I(S_n) = L$$

but was ultimately unable to prove what the value is.

- Building on Brown's equations, we will show that $L = \frac{1}{\sigma^2}$. This implies that $J_n \to 0$ and thus $D_n \to 0$.
- To do so, we need the help of an auxiliary lemma.
- The main theorem follows by showing the limit of exists and is 0 by Fekete's Lemma.
- We will then illustrate some nice corrollaries and convergence rate analysis from *Johnson and Barron (2003)* that builds on this paper.

Brown's result

To prove Lemma 2, we first extend the arguments of Brown (1982). First we state his contribution:

BROWN'S RESULT: Fix $\tau > 0$ and set $Y_k = S_{2^k} + Z_\tau$. Let $\rho_k = g_k'/g_k$ be the score function for Y_k. Note that

$$Y_{k+1} = (Y_k + Y_k')/\sqrt{2}$$

where Y_k' is an independent copy of Y_k.

To check this equation $Y_{k+1} = (Y_k + Y_k')/\sqrt{2}$, we verify this property separately for S_{2^k} and Z_τ.

Note that

$$S_{2^{k+1}} = \sum_{i=1}^{2^{k+1}} x_i/\sqrt{2^{k+1}} = \sum_{i=1}^{2^{k+1}} x_i/2^{\frac{k+1}{2}},$$

and

$$S_{2^k} + S_{2^k}' = \sum_{i=1}^{2^k} x_i/\sqrt{2^k} + \sum_{i=1}^{2^k} x_i/\sqrt{2^k}.$$

Brown's argument: independence

Because of the independence,

$$S_{2^k} + S'_{2^k} = \sum_{i=1}^{2^k} x_i/\sqrt{2^k} + \sum_{i=1}^{2^k} x_i/\sqrt{2^k} = \sum_{i=1}^{2^{k+1}} x_i/2^{k/2}.$$

Then, divide one side by $\sqrt{2}$ to get the equivalence of "S_{2^k}".

For the Z_τ part, if

$$Z_\tau^1 \sim \mathcal{N}(0,\tau), Z_\tau^2 \sim \mathcal{N}(0,\tau),$$

$$\Rightarrow (Z_\tau^1 + Z_\tau^2) \sim \mathcal{N}(0,2\tau),$$

$$\Rightarrow (Z_\tau^1 + Z_\tau^2)/\sqrt{2} \sim \mathcal{N}(0,\tau) = Z_\tau.$$

It is also proved.

Therefore,

$$\boxed{Y_{k+1} = (Y_k + Y'_k)/\sqrt{2}.}$$

Brown's argument: independence

Then, we will show

$$\boxed{\rho_{k+1}(Y_{k+1}) = E[(\rho_k(Y_k) + \rho_k(Y_k'))/\sqrt{2}|\, Y_{k+1}].}$$

Noticing: given that Y_k and Y_k' are i.i.d., The density function of $S = Y_k + Y_k'$ is obtained by convolution:

$$f_S(s) = \int g_k(s-z)g_k(z)dz.$$

Then, by $Y_{k+1} = S/\sqrt{2}$, using the formula

$$X = aZ \Rightarrow f_X(x) = \frac{1}{a} f_Z(\frac{x}{a}),$$

$$g_{k+1}(Y_{k+1}) = \sqrt{2} f_S(Y_k + Y_k') = \sqrt{2} \int g_k(Y_k + Y_k' - z)g_k(z)dz.$$

Brown's argument: independence

Let $y = Y_{k+1}$. Then, we differentiate with respect to y:

$$g'_{k+1}(y) = \sqrt{2}\int \sqrt{2}g'_k(\sqrt{2}y-z)g_k(z)dz = 2\int g'_k(\sqrt{2}y-z)g_k(z)dz.$$

Using $g'_k(x) = \rho_k(x)g_k(x)$,

$$g'_{k+1}(y) = 2\int \rho_k(\sqrt{2}y-z)g_k(\sqrt{2}y-z)g_k(z)dz.$$

Thus,

$$\rho_{k+1}(y) = \frac{g'_{k+1}(y)}{g_{k+1}(y)} = \sqrt{2}\frac{\int \rho_k(\sqrt{2}y-z)g_k(\sqrt{2}y-z)g_k(z)dz}{\int g_k(\sqrt{2}y-z)g_k(z)dz}.$$

Also, Y_k and Y'_{k+1} are symmetric in the convolution. Therefore,

$$\rho_{k+1}(y) = \sqrt{2}\frac{\int \frac{\rho_k(\sqrt{2}y-z)+\rho_k(z)}{2}g_k(\sqrt{2}y-z)g_k(z)dz}{\int g_k(\sqrt{2}y-z)g_k(z)dz},$$

Brown's argument: independence

$$\rho_{k+1}(y) = \frac{1}{\sqrt{2}} \frac{\int \Big(\rho_k(\sqrt{2}y - z) + \rho_k(z)\Big) g_k(\sqrt{2}y - z) g_k(z) dz}{\int g_k(\sqrt{2}y - z) g_k(z) dz}$$

which is exactly the conditional expectation of

$$\boxed{E[\frac{\rho_k(Y_k) + \rho_k(Y_k')}{\sqrt{2}} | Y_{k+1} = y].}$$

Brown's argument: independence

Hence, the best estimate of the sum of the scores in the sense of minimum mean-squared error (MMSE) is the score of the sum $\rho_{k+1}(Y_{k+1})$.

Conditional expectation has a very important geometric property: $E[Z|\mathcal{G}]$ is **the best squared approximation** to Z. That is

$$\boxed{E\left[(Z - E[Z|\mathcal{G}])^2\right] = \min_{h(Y_{k+1})} E\left[(Z - h(Y_{k+1}))^2\right].}$$

This yields the corresponding **pythagorean relation** (stated in the paper, or we can say: **orthogonal decomposition**) Then,

$$E[Z^2] = E\left[(E[Z|\mathcal{G}])^2\right] + E\left[(Z - E[Z|\mathcal{G}])^2\right].$$

Brown's argument: independence

Let $Z = \frac{\rho_k(Y_k)+\rho_k(Y_k')}{\sqrt{2}}, \mathcal{G} = \sigma(Y_{k+1})$.

$$E[Z^2] = E\left[(E[Z|\mathcal{G}])^2\right] + E\left[(Z - E[Z|\mathcal{G}])^2\right]$$

$$\Rightarrow E[Z^2] = E\left[\rho_{k+1}(Y_{k+1})^2\right] + E\left[(Z - \rho_{k+1}(Y_{k+1}))^2\right]$$

Then, the left-hand side simplifies as

$$E[Z^2] = E\left[\left(\frac{\rho_k(Y_k) + \rho_k(Y_k')}{\sqrt{2}}\right)^2\right]$$

$$= \frac{1}{2}E[\rho_k(Y_k)^2 + 2\rho_k(Y_k)\rho_k(Y_k') + \rho_k(Y_k')^2].$$

Brown's argument: independence

Independence and $E[\rho_k(Y_k)] = 0$ imply $E[\rho_k(Y_k)\rho_k(Y_k')] = 0$, hence

$$E[Z^2] = \tfrac{1}{2}\big(E[\rho_k(Y_k)^2] + E[\rho_k(Y_k')^2]\big) = I(Y_k).$$

Thus,

$$\boxed{I(Y_k) - I(Y_{k+1}) = E\left[\left(\frac{\rho_k(Y_k) + \rho_k(Y_k')}{\sqrt{2}} - \rho_{k+1}(Y_{k+1})\right)^2\right].}$$

Brown's argument: Show that $I(Y_k) - I(Y_{k+1}) \to 0$

This difference sequence between $I(Y_k)$ and $I(Y_{k+1})$ must **converge to zero**. Because the sequence $I(Y_k)$ is **decreasing and bounded**:

- $I(Y_k) - I(Y_{k+1}) = E(\text{ square term }) \geq 0$, which means that $I(Y_k)$ is decreasing;
- Using the formula $I(Y + V) \leq I(Y), I(Y_k) = I(S_{2^k} + Z_\tau) \leq I(Z_\tau) = 1/\tau$, which is obviously bounded.

Brown's argument:

Therefore,

$$\boxed{E\left[\left(\frac{\rho_k(Y_k)+\rho_k(Y_k')}{\sqrt{2}}-\rho_{k+1}(Y_{k+1})\right)^2\right]\to 0.}$$

Then, we want to prove

$$\boxed{\mathbb{E}\Big(\frac{\rho_k(Z_{\tau/2})+\rho_k(Z'_{\tau/2})}{\sqrt{2}}-\rho_{k+1}\big(\frac{Z_{\tau/2}+Z'_{\tau/2}}{\sqrt{2}}\big)\Big)^2\to 0.}$$

First, we want to show the density for $Y_k = S_{2^k} + Z_\tau$ is bounded below by a multiple of the normal $(0, \tau/2)$ density.

Prove the boundedness

Notice that

$$Y_k = S_{2^k} + Z_\tau,$$

we have the property for density functions

$$\rho_k(Y_k) = \rho_k(S_{2^k}) * \phi(\tau), \phi(\tau) = \phi(\tau/2) * \phi(\tau/2),$$

where $\phi(\tau)$ is density function of Z_τ. Then, we will have

$$g_k(Y_k) = g_k(S_{2^k}) * \phi(\tau) = g_k(S_{2^k}) * \phi(\tau/2) * \phi(\tau/2),$$

$$\Rightarrow \boxed{g_k(y) \geq c\phi_{\tau/2}(y), \forall y.}$$

Boundedness

Here, $\varphi_\tau(x) = (2\pi\tau)^{-1/2} e^{-x^2/(2\tau)}$.

Proof: Choose $R > 0$ and set $p = \inf_k \mathbb{P}(|S_{2^k}| \le R) > 0$ (possible since $\mathrm{Var}(S_{2^k})$ is fixed). Then,

$$g_k(y) = \int g_{S_{2^k}}(s)\, \varphi_\tau(y-s)\, ds \ge p \inf_{|s|\le R} \varphi_\tau(y-s).$$

For $|s| \le R$,

$$(y-s)^2 \le 2y^2 + 2R^2 \quad \Rightarrow \quad \varphi_\tau(y-s) \ge \tfrac{1}{\sqrt{2}} e^{-R^2/\tau}\, \varphi_{\tau/2}(y).$$

Combining the two bounds gives

$$g_k(y) \ge c_\tau\, \varphi_{\tau/2}(y), \qquad c_\tau = \tfrac{p}{\sqrt{2}} e^{-R^2/\tau} > 0.$$

□

Prove the boundedness

By using this result $g_k(y) \geq c\phi_{\tau/2}(y), \forall y$, we have

$$I(Y_k) - I(Y_{k+1}) = E\left[\left(\frac{\rho_k(Y_k) + \rho_k(Y_k')}{\sqrt{2}} - \rho_{k+1}(Y_{k+1})\right)^2\right]$$

$$\geq \frac{1}{c^2}\mathbb{E}(\frac{\rho_k(Z_{\tau/2}) + \rho_k(Z'_{\tau/2})}{\sqrt{2}} - \rho_{k+1}(\frac{Z_{\tau/2} + Z'_{\tau/2}}{\sqrt{2}}))^2.$$

In the next slide, we will discuss it in detail.

Prove the boundedness

Let $H : \mathbb{R}^2 \to [0, \infty)$ be any nonnegative measurable function. The pair (Y_k, Y_k') has density $g_k(y)g_k(y')$, so

$$\mathbb{E}\big[H(Y_k, Y_k')\big] = \iint_{\mathbb{R}^2} H(y, y')\, g_k(y)\, g_k(y')\, dy\, dy'.$$

Using $g_k(y) \geq c\phi_{\tau/2}(y)$, we have

$$g_k(y)\, g_k(y') \;\geq\; c^2\, \phi_{\tau/2}(y)\, \phi_{\tau/2}(y'), \qquad \forall\, y, y',$$

and therefore

$$\begin{aligned}
\mathbb{E}\big[H(Y_k, Y_k')\big] &= \iint H(y, y')\, g_k(y)\, g_k(y')\, dy\, dy' \\
&\geq\; c^2 \iint H(y, y')\, \phi_{\tau/2}(y)\, \phi_{\tau/2}(y')\, dy\, dy' \\
&= c^2\, \mathbb{E}\big[H(Z_{\tau/2}, Z_{\tau/2}')\big].
\end{aligned}$$

Boundedness

Equivalently,

$$\boxed{\mathbb{E}\big[H(Z_{\tau/2}, Z'_{\tau/2})\big] \;\le\; \frac{1}{c^2}\,\mathbb{E}\big[H(Y_k, Y'_k)\big].}$$

Here, we have

$$H_k(y, y') = \left(\frac{\rho_k(y) + \rho_k(y')}{\sqrt{2}} \;-\; \rho_{k+1}\left(\frac{y + y'}{\sqrt{2}}\right)\right)^2.$$

This function is nonnegative, and the Fisher–information identity gives

$$I(Y_k) - I(Y_{k+1}) = \mathbb{E}\big[H_k(Y_k, Y'_k)\big].$$

Substituting H_k into the equation above yields our desired results.

Brown's Argument:

Key point: We already have

$$\boxed{\mathbb{E}\Big(\frac{\rho_k(Z_{\tau/2})+\rho_k(Z'_{\tau/2})}{\sqrt{2}}-\rho_{k+1}\big(\frac{Z_{\tau/2}+Z'_{\tau/2}}{\sqrt{2}}\big)\Big)^2\to 0.}$$

But for an arbitrary function v with finite $\mathbb{E}v^2(Z)$,(here v is ρ_k) the mean-squared error of estimation of $v(Z)+v(Z')$ using any function (here, it's ρ_{k+1}) of the sum $Z+Z'$ is shown to exceed a multiple of the mean-squared error using the best linear estimate.

We will explain it in the next slide.

Key results

We have following property:

Fact: In a Hilbert space, the L^2-best approximation of U by functions of S is linear :

$$U_{\text{lin}}(S) = aS,$$

where a is a constant.
Key inequality. For any measurable $g(S)$,

$$\boxed{\mathbb{E}(U - g(S))^2 \;\geq\; C\,\mathbb{E}(U - U_{\text{lin}}(S))^2} \qquad \text{(A)}$$

where $C > 0$.

Application

Recursion for score function: Let ρ_k be the score of $Y_k = S_{2^k} + Z_\tau$. Define

$$U = \frac{\rho_k(Z_{\tau/2}) + \rho_k(Z'_{\tau/2})}{\sqrt{2}}, \qquad S = \frac{Z_{\tau/2} + Z'_{\tau/2}}{\sqrt{2}}.$$

The recursion gives

$$\rho_{k+1}(S) = \mathbb{E}[U \mid S]\,,$$

which is the L^2-best nonlinear estimator of U using only S.

Since,

$$\mathbb{E}(U - \rho_{k+1}(S))^2 \longrightarrow 0,$$

combine with (A):

$$\mathbb{E}(U - g(S))^2 \geq C\,\mathbb{E}(U - a_k S)^2.$$

Since the LHS $\to 0$, the RHS must also $\to 0$:

$$\boxed{\mathbb{E}(p_k(Z_{\tau/2}) - a_k Z_{\tau/2})^2 \longrightarrow 0.} \qquad \text{(B)}$$

Identification of limit

Now we have

$$p_k(z) \approx a_k z \quad \text{in } L^2(N(0, \tau/2))$$

from property (B).

Because

$$I(Y_k) \downarrow I_\infty \qquad \text{and} \qquad I_\infty = 1/(\sigma^2 + \tau),$$

the slope must satisfy

$$a_k \longrightarrow -\frac{1}{\sigma^2}.$$

Thus, the only possible limit is the Gaussian score:

$$p_\phi(z) = -\frac{z}{\sigma^2}, \qquad \phi = N(0, \sigma^2).$$

Finally, we have

$$\boxed{\mathbb{E}\left(p_k(Z_{\tau/2}) - p_\phi(Z_{\tau/2})\right)^2 \longrightarrow 0.}$$

Brown's argument:

And furthermore, since the score is the derivative of the logarithm of the density, the sequence of densities g_k for $Y_k = S_{2^k} + Z_\tau$ satisfies

$$\boxed{g_k(y) \to \phi(y)}$$

uniformly on compact subsets. Here ϕ is the normal $(0, \sigma^2 + \tau)$ density.

Brown used $g_k(y) \to \phi(y)$ and let $\tau \to 0$ to prove convergence in distribution. Next, we will use both of these equations plus a uniform integrability argument to identify the limit of the Fisher information!

Next

Let us welcome Shujun Xiong!

Lemma 2 Statement

Let $S'_n = \sqrt{t}S_n + \sqrt{1-t}Z, Z \sim \mathcal{N}(0, \sigma^2)$ for a fixed $0 \leq t < 1$.

Subadditive Sequence

The sequence $nJ(S'_n)$ is subadditive:

$$(p+q)J(S'_{p+q}) \leq pJ(S'_p) + qJ(S'_q)$$

In particular, this implies $J(S'_{2n}) \leq J(S'_n)$.

Convergence of Standardized Fisher Information

Furthermore, the *standardized Fisher information* converges to zero:

$$\lim_{n\to\infty} J(S'_n) = 0. \tag{1}$$

Equivalently, the sequence of Fisher informations $I(S'_n) \to 1/\sigma^2$.

Lemma 2: Overview of Arguments

First, consider the primary object of study

$$Y_k = S_{2^k} + Z_\tau, Z_\tau = \mathcal{N}(0, \tau), S_{2^k} = \frac{1}{\sqrt{2^k}} \sum_{i=1}^{2^k} X_i$$

Lemma 2: Overview of Arguments

First, consider the primary object of study

$$Y_k = S_{2^k} + Z_\tau, Z_\tau = \mathcal{N}(0, \tau), S_{2^k} = \frac{1}{\sqrt{2^k}} \sum_{i=1}^{2^k} X_i$$

$$I(Y_k) = \mathbb{E}(\rho_k(Y_k)^2) = \int \rho_k^2 \frac{g_k}{\phi} d\Phi, \Phi = \mathcal{N}(0, \sigma^2 + \tau)$$

Lemma 2: Overview of Arguments

First, consider the primary object of study

$$Y_k = S_{2^k} + Z_\tau, Z_\tau = \mathcal{N}(0, \tau), S_{2^k} = \frac{1}{\sqrt{2^k}} \sum_{i=1}^{2^k} X_i$$

$$I(Y_k) = \mathbb{E}(\rho_k(Y_k)^2) = \int \rho_k^2 \frac{g_k}{\phi} d\Phi, \Phi = \mathcal{N}(0, \sigma^2 + \tau)$$

We denote the density of Y by g_k, the score function by $\rho_k = \frac{g_k'}{g_k}$.

Lemma 2: Overview of Arguments

1. $\rho_k \to \rho_\phi$ in the associated Gaussian distribution.

Lemma 2: Overview of Arguments

1. $\rho_k \to \rho_\phi$ in the associated Gaussian distribution.
2. $\frac{g_k}{\phi} \to 1$ in Φ probability.

Lemma 2: Overview of Arguments

1. $\rho_k \to \rho_\phi$ in the associated Gaussian distribution.
2. $\frac{g_k}{\phi} \to 1$ in Φ probability.
3. Show that $\lim_{k\to\infty} \int \rho_k^2 \frac{g_k}{\phi} d\Phi = \int \rho_\phi^2 d\Phi$

Lemma 2: Overview of Arguments

1. $\rho_k \to \rho_\phi$ in the associated Gaussian distribution.
2. $\frac{g_k}{\phi} \to 1$ in Φ probability.
3. Show that $\lim_{k\to\infty} \int \rho_k^2 \frac{g_k}{\phi} d\Phi = \int \rho_\phi^2 d\Phi$
4. Show uniform Φ-integrability. This is the key advancement made by this paper to improve on prior work.

Lemma 2: Overview of Arguments

1. $\rho_k \to \rho_\phi$ in the associated Gaussian distribution.
2. $\frac{g_k}{\phi} \to 1$ in Φ probability.
3. Show that $\lim_{k\to\infty} \int \rho_k^2 \frac{g_k}{\phi} d\Phi = \int \rho_\phi^2 d\Phi$
4. Show uniform Φ-integrability. This is the key advancement made by this paper to improve on prior work.
5. $\lim_{k\to\infty} I(S_{2^k}) = \frac{1}{\sigma^2}$

Lemma 2: Overview of Arguments

1. $\rho_k \to \rho_\phi$ in the associated Gaussian distribution.
2. $\frac{g_k}{\phi} \to 1$ in Φ probability.
3. Show that $\lim_{k\to\infty} \int \rho_k^2 \frac{g_k}{\phi} d\Phi = \int \rho_\phi^2 d\Phi$
4. Show uniform Φ-integrability. This is the key advancement made by this paper to improve on prior work.
5. $\lim_{k\to\infty} I(S_{2^k}) = \frac{1}{\sigma^2}$
6. Apply Fekete's Lemma on sub-additivity to get the desired limit.

Lemma 2: Setting up uniform convergence

1. ASSUME: $\rho_k \to \rho_\phi$ in the associated Gaussian distribution.
2. ASSUME: $\frac{g_k}{\phi} \to 1$ in Φ probability.
3. Show that $\lim_{k\to\infty} \int \rho_k^2 \frac{g_k}{\phi} d\Phi = \int \rho_\phi^2 d\Phi$

Lemma 2: Show that $\lim_{k\to\infty} \int \rho_k^2 \frac{g_k}{\phi} d\Phi = \int \rho_\phi^2 d\Phi$

First, by the continuous mapping theorem, the product of sequences convergent in probability is also convergent in probability.

Lemma 2: Show that $\lim_{k\to\infty} \int \rho_k^2 \frac{g_k}{\phi} d\Phi = \int \rho_\phi^2 d\Phi$

First, by the continuous mapping theorem, the product of sequences convergent in probability is also convergent in probability.

This is nice because we have assumed

- $\lim_{k\to\infty} \rho_k^2 \to \rho_\phi^2$
- $\lim_{k\to\infty} \frac{g_k}{\phi} \to 1$

Thus, we get

$$\lim_{k\to\infty} \rho_k^2 \frac{g_k}{\phi} = \rho_\phi^2 \cdot 1 = \rho_\phi^2.$$

We would like to swap the order of the limit and the integral. A sufficient condition is uniform Φ-integrability.

Lemma 2: Showing uniform Φ-integrability

Our argument rests on a standard proof that bounded relative entropy implies uniform integrability of the density ratio.
In an auxiliary lemma, we found that $\rho_k^2 \frac{g_k}{\phi} \leq C \frac{g_{k,2\tau}}{\phi}$ (proof omitted for time).

Lemma 2: Showing uniform Φ-integrability

Our argument rests on a standard proof that bounded relative entropy implies uniform integrability of the density ratio.
In an auxiliary lemma, we found that $\rho_k^2 \frac{g_k}{\phi} \leq C \frac{g_{k,2\tau}}{\phi}$ (proof omitted for time).
To show boundedness, consider relative entropy

$$\int \frac{g_{k,2\tau}}{\phi} \log \frac{g_{k,2\tau}}{\phi} d\Phi$$

This is just the KL-divergence between Y_k and Φ, so there is a lower bound, and we only need to show that there is an upper bound.

Lemma 2: Showing uniform Φ-integrability

Because $-H(S_{2^k} + Z_{2\tau}) \leq -H(Z_{2\tau})$, then $g_{k,2\tau} \log g_{k,2\tau} \leq C$, or rather it is bounded by a constant $H(Z_{2\tau})$. This means that we can simplify the integral as follows:

$$\begin{aligned}
\int \frac{g_{k,2\tau}}{\phi} \log \frac{g_{k,2\tau}}{\phi} d\Phi &= \int \frac{g_{k,2\tau}}{\phi} \log g_{k,2\tau} - \frac{g_{k,2\tau}}{\phi} \log \frac{1}{\phi} d\Phi \\
&\leq \int \frac{C + g_{k,2\tau}(a)(\frac{1}{2}\log(2\pi\sigma^2) + \frac{a^2}{2\sigma^2})}{\phi(a)} d\Phi(a) \\
&= C + \int_{\mathbb{R}} g_{k,2\tau}(a)(\frac{1}{2}\log(2\pi\sigma^2) + \frac{a^2}{2\sigma^2}) da \\
&\leq C + \frac{1}{2\sigma^2} + \int g_{k,2\tau}(a) a^2 da.
\end{aligned}$$

The integral in the last line corresponds to variance of Y_k, which we know is finite.

Lemma 2: $\lim_{k\to\infty} I(S_{2^k}) = \frac{1}{\sigma^2}$

We know that

$$\begin{aligned} I(Y_k) &= I(S_{2^k} + \sqrt{\tau}Z) \\ &= I(\frac{1}{\sqrt{t}}(\sqrt{t}S_{2^k} + \sqrt{1-t}Z)), \tau = \frac{1-t}{t} \\ &= \frac{1}{t} I(S'_{2^k}) \end{aligned}$$

Lemma 2: $\lim_{k\to\infty} I(S_{2^k}) = \frac{1}{\sigma^2}$

We know that

$$\begin{aligned} I(Y_k) &= I(S_{2^k} + \sqrt{\tau}Z) \\ &= I(\frac{1}{\sqrt{t}}(\sqrt{t}S_{2^k} + \sqrt{1-t}Z)), \tau = \frac{1-t}{t} \\ &= \frac{1}{t} I(S'_{2^k}) \end{aligned}$$

Thus,

$$\lim_{k\to\infty} I(Y_k) = \lim_{k\to\infty} \int \rho_k^2 \frac{g_k}{\phi} d\Phi = \int \rho_\phi^2 d\Phi = \frac{1}{\sigma^2 + \tau}$$

$$\lim_{k\to\infty} I(S'_{2^k}) = \frac{t^2}{t\sigma^2 + 1 - t}$$

Taking limit of $t \to 1$ gives the desired bound.

Lemma 2: Apply sub-additivity to finish the proof

Fekete's Lemma says that for any sub-additive sequence a_n, the limit

$$\lim_{n\to\infty} \frac{a_n}{n}$$

exists and equals $\inf_{k\geq 1} \frac{a_k}{k}$. This means that the entire sequence has the same limit as the subsequence we studied, so

$$\lim_{k\to\infty} I(S_{2^k}) = \lim_{k\to\infty} I(S_k) = \frac{1}{\sigma^2}.$$

Entropic CLT Statement

Theorem

Let S_n be the standardized sum. Then $nD(S_n)$ is a subadditive sequence. In particular, $D(S_{2n}) \leq D(S_n)$. Furthermore, the relative

entropy converges to zero

$$\lim_{n \to \infty} D(S_n) = 0 \tag{2}$$

if and only if $D(S_n)$ is finite for some n.

Proof of limit

Recall Lemma 1:

$$D(X) = \int_0^1 J(\sqrt{t}X + \sqrt{1-t}Z)\frac{dt}{2t},$$

Duality between D and J makes this simple to see. Consider the subsequence $n_k = 2^k$.

Proof of limit

Recall Lemma 1:

$$D(X) = \int_0^1 J(\sqrt{t}X + \sqrt{1-t}Z)\frac{dt}{2t},$$

Duality between D and J makes this simple to see. Consider the subsequence $n_k = 2^k$.
Lemma 2 gives $J(S'_{n_k}) \downarrow 0$. For $D(S_n)$ finite for some n, monotone convergence theorem ($D(S_n)$ is monotone decreasing and bounded), then $D(S_{n_k}) \downarrow 0$.
The entire sequence has the same limit as the subsequence, by a similar argument as in Lemma 2, so

$$\lim_{n\to\infty} D(S_n) = 0$$

Proof of limit

Recall Lemma 1:

$$D(X) = \int_0^1 J(\sqrt{t}X + \sqrt{1-t}Z)\frac{dt}{2t},$$

Duality between D and J makes this simple to see. Consider the subsequence $n_k = 2^k$.
Lemma 2 gives $J(S'_{n_k}) \downarrow 0$. For $D(S_n)$ finite for some n, monotone convergence theorem ($D(S_n)$ is monotone decreasing and bounded), then $D(S_{n_k}) \downarrow 0$.
The entire sequence has the same limit as the subsequence, by a similar argument as in Lemma 2, so

$$\lim_{n\to\infty} D(S_n) = 0$$

The converse follows naturally; for the limit of the sequence to be 0, then it needs to be finite at some point in the sequence.

Corollaries

Assume $H(S_n)$ is finite for some n. Then, S_n has a density function f_n which converges to ϕ in L^1 and the difference between $\log f_n(S_n)$ and $\log \phi(S_n)$ converges in probability as well. More formally,

- $\lim_{n\to\infty} \int |f_n(x) - \phi(x)| dx = 0$
- $\lim_{n\to\infty} \mathbb{E}(|f_n(S_n) - \phi(S_n)|) = 0$

Proof of Corollaries

- By information theory bounds between D_{KL} and TV, $2D(f) \geq (\int |f - \phi|)^2 \implies \int |f - \phi| \leq \sqrt{2D}$. Since $D := D(S_n) \to 0$, then the LHS $\to 0$ as well.

Proof of Corollaries

- By information theory bounds between D_{KL} and TV, $2D(f) \geq (\int |f - \phi|)^2 \implies \int |f - \phi| \leq \sqrt{2D}$. Since $D := D(S_n) \to 0$, then the LHS $\to 0$ as well.
- Next, consider $\mathbb{E}(|\log f_n(S_n) - \log \phi(S_n)|) = \int f_n(S_n) |\log \frac{f_n(S_n)}{\phi(S_n)}|$. We endeavor to show that $D(f) + \sqrt{2D(f)} \geq \int f |\log \frac{f}{\phi}|$.

Proof of Corollaries (cont.)

Consider the set $A := \{x : \phi(x) > f(x)\}$.

$$\int f|log\frac{f}{\phi}| = \int_A f \log \frac{\phi}{f} + \int_{A^{\complement}} f \log \frac{f}{\phi}.$$

Proof of Corollaries (cont.)

Consider the set $A := \{x : \phi(x) > f(x)\}$.

$$\int f|log\frac{f}{\phi}| = \int_A f \log \frac{\phi}{f} + \int_{A^{\complement}} f \log \frac{f}{\phi}.$$

$$\int_{A^{\complement}} f \log \frac{f}{\phi} = D - \int_A f \log \frac{f}{\phi} = D + \int_A f \log \frac{\phi}{f}$$

Proof of Corollaries (cont.)

Consider the set $A := \{x : \phi(x) > f(x)\}$.

$$\int f|log\frac{f}{\phi}| = \int_A f \log \frac{\phi}{f} + \int_{A^\complement} f \log \frac{f}{\phi}.$$

$$\int_{A^\complement} f \log \frac{f}{\phi} = D - \int_A f \log \frac{f}{\phi} = D + \int_A f \log \frac{\phi}{f}$$

$$\int_A f \log \frac{\phi}{f} \leq \frac{1}{2} \int |f - \phi|, \text{since} \log \frac{x}{y} \leq \frac{x}{y} - 1$$

Adding the terms together shows that the upper bound is $D + \int |f - \phi| \leq D + \sqrt{2D}$.

Johnson and Barron (2004): Rate of Convergence

Chief idea is to study constraints on the score function ρ.

- Regularity condition called *weak differentiability* attains nice decomposition of the expectation of the derivative of a density function.
- Use Poincare inequalities to bound the derivative of the density function to its value (i.e. the score function)
- Apply analysis from Brown about the squared distance between the score functions to get the desired bound.

This method also improves generality of the bounds for non-perturbative constructions; in other words, these results hold for random variables X, as opposed to constructing $Y = X + Z_\tau$.

Final Remarks

Take-home message

- **Classic CLT**: standardized sums converge in distribution to the Gaussian.
- **Our result**: under a mild regularity assumption, this convergence **also holds in entropy**.

Thank you for your attention!

Questions are welcome.

Auxiliary Lemma

This lemma is necessary for the uniform integrability argument vital to the core Lemma of this paper (Lemma 2).
Let g_τ be the density for $Y = X + Z_\tau$ for an arbitrary random variable X and $Z_\tau \sim \mathcal{N}(0, \tau)$. Then, we have

$$g'_\tau(y)^2 \leq c_\tau g_{2\tau}(y) g_\tau(y), c_\tau = \frac{4\sqrt{2}}{e\tau}.$$

Auxiliary Lemma: Show that $g_\tau(y) = \mathbb{E}(\phi_\tau(y - X))$

Consider $p :=$ pdf of X. Since X, Z are independent, g_τ, the pdf of Y, is a convolution of their pdfs.

$$g_\tau(y) = (p * y)(y) = \int_{-\infty}^{\infty} p(x)\phi_\tau(y - x)dx \tag{3}$$

$$= \mathbb{E}(\phi_\tau(y - X)). \tag{4}$$

Auxiliary Lemma: Show that $g'_\tau(y) = \mathbb{E}(\phi'_\tau(y - X))$

We know that $\phi'_\tau(z) = -z\frac{\phi_\tau(z)}{\tau}$.

$$g'_\tau(y) = \lim_{\epsilon \to 0} \frac{g_\tau(y+\epsilon) - g_\tau(y)}{\epsilon} \tag{5}$$

$$= \lim_{\epsilon \to 0} \mathbb{E}\left(\frac{\phi_\tau(y+\epsilon-X) - \phi_\tau(y-X)}{\epsilon}\right). \tag{6}$$

By MVT, $\forall \epsilon, \exists r.v. C_\epsilon(X) \in (y - X, y + \epsilon - X) s.t.$

$$\frac{\phi_\tau(y+\epsilon-X) - \phi_\tau(y-X)}{\epsilon} = \phi'_\tau(C_\epsilon(X)).$$

Auxiliary Lemma: Show that $g'_\tau(y) = \mathbb{E}(\phi'_\tau(y - X))$

Now, we apply the Bounded Convergence Theorem: if a sequence of r.v.s is dominated by an integrable r.v., then the limit of their expectation = the expectation of their limit.

- Consider the sequence $U_\epsilon = \frac{\phi_\tau(y+\epsilon-X)-\phi_\tau(y-X)}{\epsilon} = \phi'_\tau(C_\epsilon(X))$.
- Note that ϕ'_τ is bounded and continuous (since it is a constant multiple of the normal pdf) and is dominated by an integrable function (any larger multiple).
- Thus, BCT applies.

With this, we get

$$\lim_{\epsilon \to 0} \mathbb{E}(U_\epsilon) = \mathbb{E}(\lim_{\epsilon \to 0} U_\epsilon) = \mathbb{E}(\phi'_\tau(y - X)).$$

Auxiliary Lemma: Show that $g'_\tau(y)^2 \leq c_\tau \mathbb{E}(\phi_{2\tau}(y - X)g_\tau(y))$

First, apply Cauchy-Schwarz,

$$g'_\tau(y)^2 = (\mathbb{E}(\frac{1}{\tau}(y - X))\phi_\tau(y - X)))^2 \tag{7}$$

$$= \frac{1}{\tau^2}(\int_{-\infty}^{\infty}(y - x)(\sqrt{\phi_\tau(y - x)p(x)})^2 dx)^2 \tag{8}$$

$$\leq (\int_{-\infty}^{\infty}(y - x)^2\phi_\tau(y - x)p(x))(\int_{-\infty}^{\infty}\phi_\tau(y - x)p(x)dx) \tag{9}$$

$$\leq \frac{1}{\tau^2}\mathbb{E}((y - X)^2\phi_\tau(y - x))g_\tau(y). \tag{10}$$

By the first claim, we know that $\mathbb{E}(\phi_{2\tau}(y - X)) = g_{2\tau}(y)$, so the proof is completed by substituting into

$$c_\tau \mathbb{E}(\phi_{2\tau}(y - X)g_\tau(y)) = c_\tau g_{2\tau}(y)g_\tau(y).$$

Bounded relative entropy implies uniform integrability of the density ratio

Consider $W = \frac{g_{k,\tau}}{\phi}$. Clearly, $D_{KL}(Y_k, \Phi) = \mathbb{E}_{\Phi}(W \log W)$. Because the KL-divergence is bounded, then this expectation is bounded as well.

For uniform integrability, we just need to ensure that

$$\lim_{\delta \to \infty} \mathbb{E}(W \mathbf{1}(W > \delta)) \to 0.$$

Bounded relative entropy implies uniform integrability of the density ratio

$$\mathbb{E}(W\mathbf{1}(W > \delta)) \leq \mathbb{E}(\frac{W \log W}{\log \delta}\mathbf{1}(W > \delta))$$
$$\leq \frac{1}{\log \delta}\mathbb{E}(W \log W).$$

The expectation is bounded, so as $\delta \to \infty$, then this bound goes to 0, and we are satisfied with uniform integrability!
Thus, $\frac{g_{k,2\tau}}{\phi}$ is uniformly Φ-integrable, so we can swap the limit with the integral and

$$\lim_{k\to\infty} \int \rho_k^2 \frac{g_k}{\phi} d\Phi = \int \rho_\phi^2 d\Phi.$$